

What we're doing

For the cleaned native + MPNN-redesigned contacts-v1 corpora from #266/#274, what model size is compute/data efficient? In particular, do 0.7B, 1.5B, and 3B Qwen-style contacts-v1 models show the expected Chinchilla-style tradeoff between parameter count, token budget, validation cross-entropy, and downstream contact R-precision?

Why

The new ~248B-token corpus is large enough that the current 1.5B recipe may be under-sized at fixed token budget. A 3B model should improve validation CE and possibly FoldBench R-precision over 1.5B if the extra MPNN/native data carries independent signal; a 0.7B model provides a cheap lower anchor for fitting a scaling trend rather than only comparing two large runs.

Planned Chinchilla measurements

Three fixed-recipe arms: 20×1536 ($\sim 0.701\text{B}$), 24×2048 ($\sim 1.471\text{B}$), and 32×2560 ($\sim 3.035\text{B}$). One packed epoch is $\sim 279.3\text{B}$ consumed tokens, giving ~ 398 , ~ 190 , and ~ 92 tokens/parameter.

For each arm we will track validation CE vs tokens, validation CE vs GPU-hours, final token/parameter ratio, throughput, and selected exp245 eval-val contact metrics.

The goal is to collect scaling evidence for the next MarInFold recipe. A smaller arm anchors the slope; the current 1.5B scale preserves continuity; the 3B arm tests whether the cleaned/redesigned corpus makes 1.5B under-sized.

Results so far

Training harness added. The four CoreWeave token caches were verified by ``/zack/exp288-prepare-a01``, so the sweep does not require a new raw-data transfer. H100s were full, so the active target is the recent successful GB200/B200 profile on ``cw-us-east-08a``: 8 nodes × 4 GB200 GPUs per trial.